

Musthofa Joko Anggoro

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Education

Universitas Indonesia, Depok

Sarjana Komputer/Bachelor of Computer Science

2022 – 2026

GPA : 3.85/4.00

English Proficiency

IELTS Academic

Overall Band 7.5 (Listening 8.0, Reading 8.5, Writing 6.5, Speaking 6.0)

July 2026

CEFR C1

Undergraduate Thesis

Bridging the Gap Between the Mano Register-Transfer Model and FPGA Realization Through a Synthesisable VHDL-93 Architecture

Supervisor:

Prof. Dr. Ir. Petrus Mursanto, M.Sc.

- Resolved undefined FPGA-synthesis constraints in the Mano register-transfer model, including register organisation, instruction encoding, and the I/O interface, into a complete 8-bit datapath in VHDL-93 on a Xilinx Spartan-3AN board
- Verified the design via ISim and GHDL simulation plus board-level LCD testing, reaching a 17.051 ns minimum clock period (58.648 MHz), clearing the 50 MHz board constraint

Publications

Musthofa Joko Anggoro and Petrus Mursanto.

Under review

"Bridging the Gap Between the Mano Register-Transfer Model and FPGA Realization Through a Synthesisable VHDL-93 Architecture." Submitted to the International Conference on Advanced Computer Science and Information Systems (ICACSIS), 2026.

Research Interests

Computer architecture, digital hardware design, FPGA, IoT and cyber-physical systems, embedded systems, smart buildings and energy sensing

Research Projects

IoT-based Modular Platform for Real-time Energy and Space Sensing 2026 – present

Principal Investigator:

Aprinaldi, S.Kom., M.Kom., Ph.D.Eng.

- Funding proposal submitted for a four-layer IoT platform for occupancy-based air conditioning control
- Own the device side: sensor nodes for temperature, humidity, occupancy, and air quality, a Zigbee2MQTT gateway, and relay-based actuation, feeding a self-hosted Mosquitto and Home Assistant stack
- Validated the prototype at TRL 4 in the lab; the submitted proposal targets a pilot deployment in one faculty room to advance to TRL 5

Work Experience

Traveloka

Aug 2025 – Jul 2026

Software Engineer Intern

- Owned the booking process for a high-traffic production system, maintaining and fixing bugs while accounting for system load and capacity constraints
- Built and improved an internal credential management system to reduce fraud
- Proposed a new framework for automating the reschedule process; adopted after team review, cutting turnaround from 2 months to 2 weeks and raising success rate from 60% to about 85%

Leadership Experience

COMPFEST, The Biggest IT Event in Indonesia, run by students Jan 2023 - Dec 2023

Vice Person in Charge of UI/UX

- Ran 5 design sprints and shipped dark and light mode for compfest.id
- Managed a team of 11 designers through a landing page and design system overhaul
- Worked with engineers to align the design system with technical constraints

Scholarships and Academic Programs

INSPIRASI-NTU Summer Program 2025

LPDP Scholarship Awardee, Nanyang Technological University, Singapore

Skills

Hardware/HDL: VHDL-93, FPGA (Xilinx Spartan-3AN), ISim, GHDL, ESP32 (ESP-IDF / Arduino framework)

IoT and embedded systems: ESP32, MQTT (Mosquitto), Zigbee2MQTT, Home Assistant, sensor integration, relay-based actuation

Tech stack: Java (Spring Boot), Redis, TypeScript, MongoDB, Terraform (AWS)